

Unit – 5 : Session – 1 : SLO – 1 | Finite Automata — Definition and Types (DFA, NFA)

I. Questions & Answers

1. What is a finite automaton? — A theoretical machine with a finite set of states that reads an input string symbol by symbol, transitions between states according to a transition function, and accepts or rejects the string depending on whether it ends in an accepting (final) state.
2. What distinguishes a DFA from an NFA? — In a DFA every state has exactly one transition per input symbol (deterministic); in an NFA a state may have multiple transitions for the same symbol, none at all, or ϵ -transitions that move without consuming input.
3. Can every NFA be converted to a DFA? — Yes. The subset construction method converts any NFA with n states into an equivalent DFA (with up to 2^n states) that recognizes exactly the same language.